

Capstone Project Documentation AI Camera System

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Contents

1	Project Overview	2
1.1	Project Description	2
1.2	My Role	2
2	Semester 1 Contributions	2
2.1	Model Selection and Evaluation	2
2.2	Edge AI Hardware Research	3
2.3	Inference Pipeline Prototype Development	3
2.4	System Integration Design	3
2.5	Preliminary Performance Validation	4
3	Semester 2 Contributions	4
3.1	Investigation of the NPU Software Stack	4
3.2	Challenges Encountered	4

1 Project Overview

1.1 Project Description

The goal of this capstone project was to develop an AI-powered camera system capable of performing real-time object detection on embedded hardware.

The system integrates:

- Camera input pipeline
- Neural network inference engine
- Object detection models (YOLO)
- Hardware acceleration using NPU & GPU
- Visualization and deployment framework

1.2 My Role

I was primarily responsible for the AI inference stack, including:

- YOLO model selection and evaluation
- Model conversion and optimization
- NPU deployment on NXP i.MX8M Plus
- GPU deployment on Jetson
- Docker-based deployment workflow
- Performance testing and debugging

2 Semester 1 Contributions

2.1 Model Selection and Evaluation

My primary responsibility during the first semester was the development of the AI inference subsystem for the camera platform. Since the system was required to perform real-time object detection on live video streams, low inference latency and high processing efficiency were identified as the most critical requirements.

I conducted an evaluation of several state-of-the-art object detection models and selected the YOLO family as the primary candidate due to its strong balance between detection accuracy and computational efficiency. Particular attention was given to lightweight variants optimized for edge deployment. Through benchmarking and literature review, YOLO26 was identified as a promising architecture because of its improved inference speed, reduced computational complexity, and lower dependency on high-end hardware accelerators.

2.2 Edge AI Hardware Research

In parallel with model evaluation, I investigated multiple hardware platforms capable of supporting edge AI workloads.

The study included:

- NVIDIA Jetson GPU platforms
- Embedded NPUs
- CPU-only deployment scenarios
- Future support for vision-language and online learning workloads

While Jetson platforms demonstrated excellent inference performance through TensorRT optimization, the project’s design constraints prioritized low power consumption and battery-powered operation. Based on performance-per-watt analysis, integrated NPUs were determined to be the most suitable processing units for real-time object detection and tracking tasks.

This investigation also provided insight into the suitability of different hardware architectures for emerging AI applications, including vision-language models, generative models, and online learning systems.

2.3 Inference Pipeline Prototype Development

To validate the feasibility of the proposed system architecture, I developed an initial inference pipeline using GStreamer.

The prototype pipeline consisted of:

1. Video frame acquisition from camera sources
2. Frame preprocessing
3. YOLO-based object detection
4. Extraction of detection metadata
5. Result transmission to backend services

For each captured frame, the pipeline generated both the raw image data and structured detection results, including object classes, confidence scores, and bounding box coordinates.

2.4 System Integration Design

I contributed to the design of the end-to-end AI camera architecture by defining the data flow between the camera subsystem, inference engine, server backend, and client dashboard.

The resulting architecture enabled:

- Real-time video processing
- Centralized inference execution
- Efficient metadata transmission
- Live visualization of detection results

This architecture established the foundation for subsequent deployment and optimization work performed during the second semester.

2.5 Preliminary Performance Validation

Performance testing was conducted on the prototype inference pipeline to evaluate real-time feasibility.

Experimental results demonstrated an average inference latency of approximately 20 milliseconds per frame, indicating that the selected architecture could support real-time video analytics requirements. These findings provided the technical justification for continuing development of the AI camera platform in the second semester.

3 Semester 2 Contributions

3.1 Investigation of the NPU Software Stack

During the second semester, the focus of the project shifted from platform familiarization to implementing an end-to-end object detection pipeline on the NXP i.MX 8M Plus embedded AI platform. The first stage involved investigating the hardware and software architecture of the target system. Particular attention was given to understanding the Neural Processing Unit (NPU), its acceleration capabilities, and the software components required to access it. This included studying the Board Support Package (BSP), TensorFlow Lite runtime, and the VX Delegate used to offload inference workloads from the CPU to the NPU.

Following the platform investigation, work was carried out on model preparation and deployment. Several YOLO-based object detection models were evaluated, and a quantized TensorFlow Lite model was generated to satisfy the NPU runtime requirements. This process required understanding model conversion, quantization techniques, and compatibility constraints imposed by the embedded hardware.

A Python-based inference application was then developed. The application loads the quantized model through TensorFlow Lite, initializes the VX Delegate, and performs object detection inference using the NPU. Additional functionality was implemented to handle image preprocessing, output tensor dequantization, non-maximum suppression, object classification, and bounding box visualization.

To evaluate the deployment, a local video file was used as the input source. Individual video frames were extracted, preprocessed, and submitted to the NPU for inference. Detection results were then post-processed and rendered onto the original frames. Performance measurements were collected for preprocessing, inference, and post-processing stages to better understand system behavior and identify potential bottlenecks.

The completed implementation successfully demonstrated hardware-accelerated object detection on the NXP i.MX 8M Plus platform. The system was able to load a quantized YOLO model, utilize the NPU through the VX Delegate, process video streams, detect objects within each frame, and generate annotated detection outputs. This achievement verified the functionality of the embedded AI deployment pipeline and established a foundation for future work involving live camera input, tracking algorithms, and real-time optimization.

3.2 Challenges Encountered

Several challenges were encountered throughout the development process. The most significant difficulties were related to configuring the software environment required for NPU acceleration. Considerable effort was spent verifying BSP support, locating the appropriate delegate libraries, and ensuring compatibility between the quantized model and the runtime environment.

Additional challenges arose from the containerized Torizon OS workflow, where software dependencies and hardware access needed to be correctly configured within Docker containers. Furthermore, validating whether inference was actually executed on the NPU rather than the CPU required extensive testing and performance analysis.

Despite these challenges, a working NPU-accelerated object detection pipeline was successfully implemented by the end of the semester.